

Abstract

Wireless Sensor Networks (WSNs) deployed for disaster monitoring face three challenges: (i) limited energy, (ii) full-lifecycle carbon footprint, and (iii) vulnerability of AI-learned policies to adversarial attacks. This thesis proposes three complementary frameworks forming a **complete protocol stack**:

GAPF — lightweight MARL policy fusion via GCN (**1,473 params**, $\text{PDR} \geq 98\%$, $2\text{--}3.5\times$ less energy);

CALASH — carbon-aware 4-pillar protocol with Lyapunov guarantees ($\text{LCI} -33\%$, lifetime $+60\%$);

CASTER-ZT — zero-trust shielded decision control (0% FP, 3.07 ms/decision, 10 formal propositions).

Keywords: WSN, MARL, GNN, policy fusion, lifecycle carbon, zero-trust, 6G, disaster monitoring.

Motivation & Research Questions

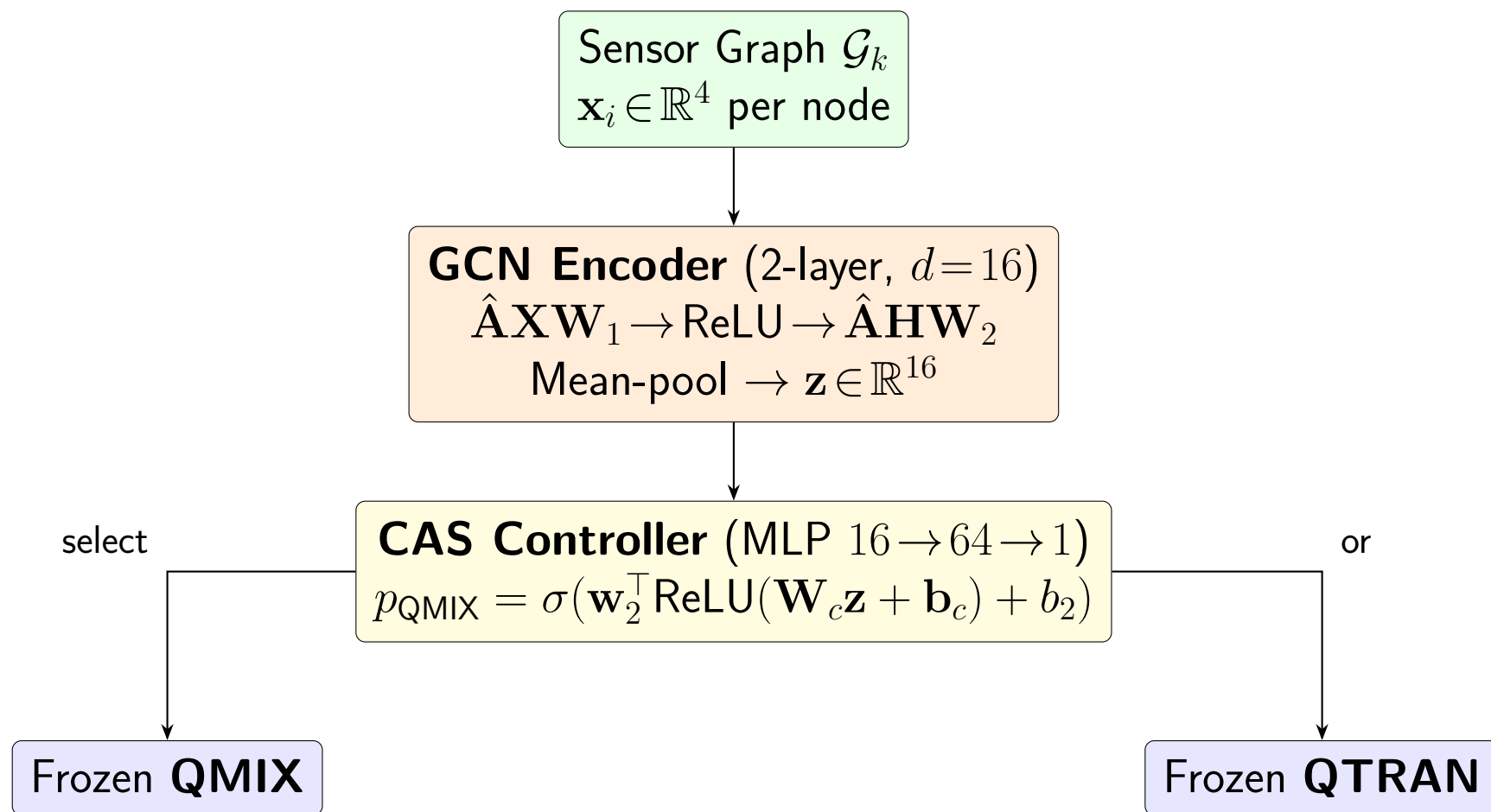
- 7,348 major disasters (2000–2019); climate change increases frequency $+30\%$.
- WSNs: non-rechargeable batteries, limited RAM (16–256 KB), physical vulnerability.
- Existing solutions address at most **4/12 design criteria** individually.

Three Specific Research Questions

- Q1** Can a lightweight graph-neural-network controller fuse pre-trained MARL policies to achieve a packet delivery ratio above 98% with fewer than 2,000 trainable parameters?
- Q2** To what extent can a four-pillar carbon-aware protocol, combined with 6G dual-band physical-layer capabilities, reduce the lifecycle carbon intensity of a disaster-monitoring WSN?
- Q3** Is a zero-trust decision shield able to detect rogue AI policies with zero false positives and enforce graduated responses within the 10 ms O-RAN real-time budget?

Article 1: GAPF — Graph-Attentive Policy Fusion

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Trainable Parameters

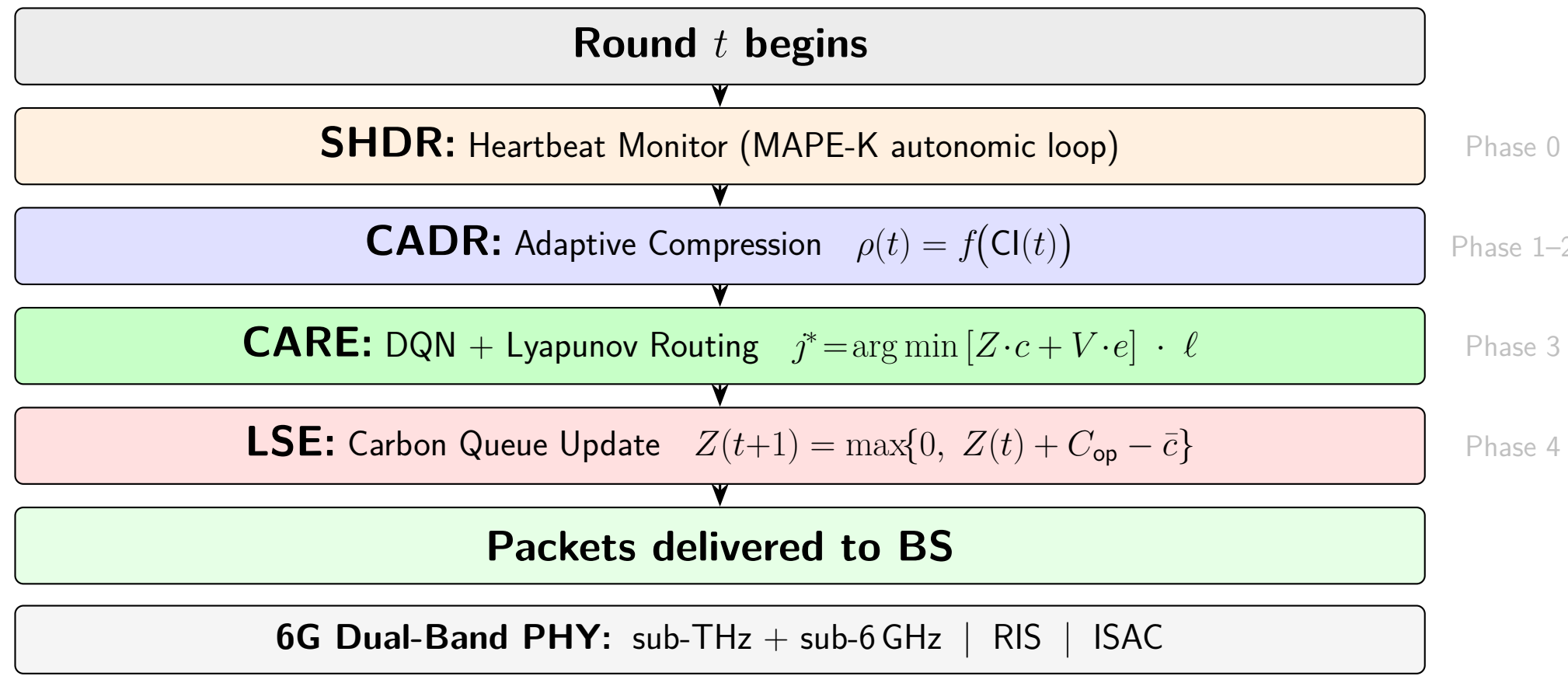
$$|\theta| = \underbrace{4 \times 16 + 16 \times 16}_{\text{GCN: } 320} + \underbrace{16 \times 64 + 64 + 64 \times 1 + 1}_{\text{CAS: } 1,153} = \mathbf{1,473}$$

Bi-Level Training

- Inner loop:** MARL experts (QMIX/QTRAN) via EPyMARL; $\alpha = 3 \times 10^{-4}$, ϵ : $0.5 \rightarrow 0.01$ (150K steps), replay 15K.
- Outer loop:** ES tunes Q-K attention ($\sigma_{\text{ES}} = 0.01$); REINFORCE trains GCN+CAS ($\alpha = 3 \times 10^{-3}$, $\beta = 0.95$).

Article 2: CALASH — Carbon-Aware Lifecycle-Autonomous Self-Healing

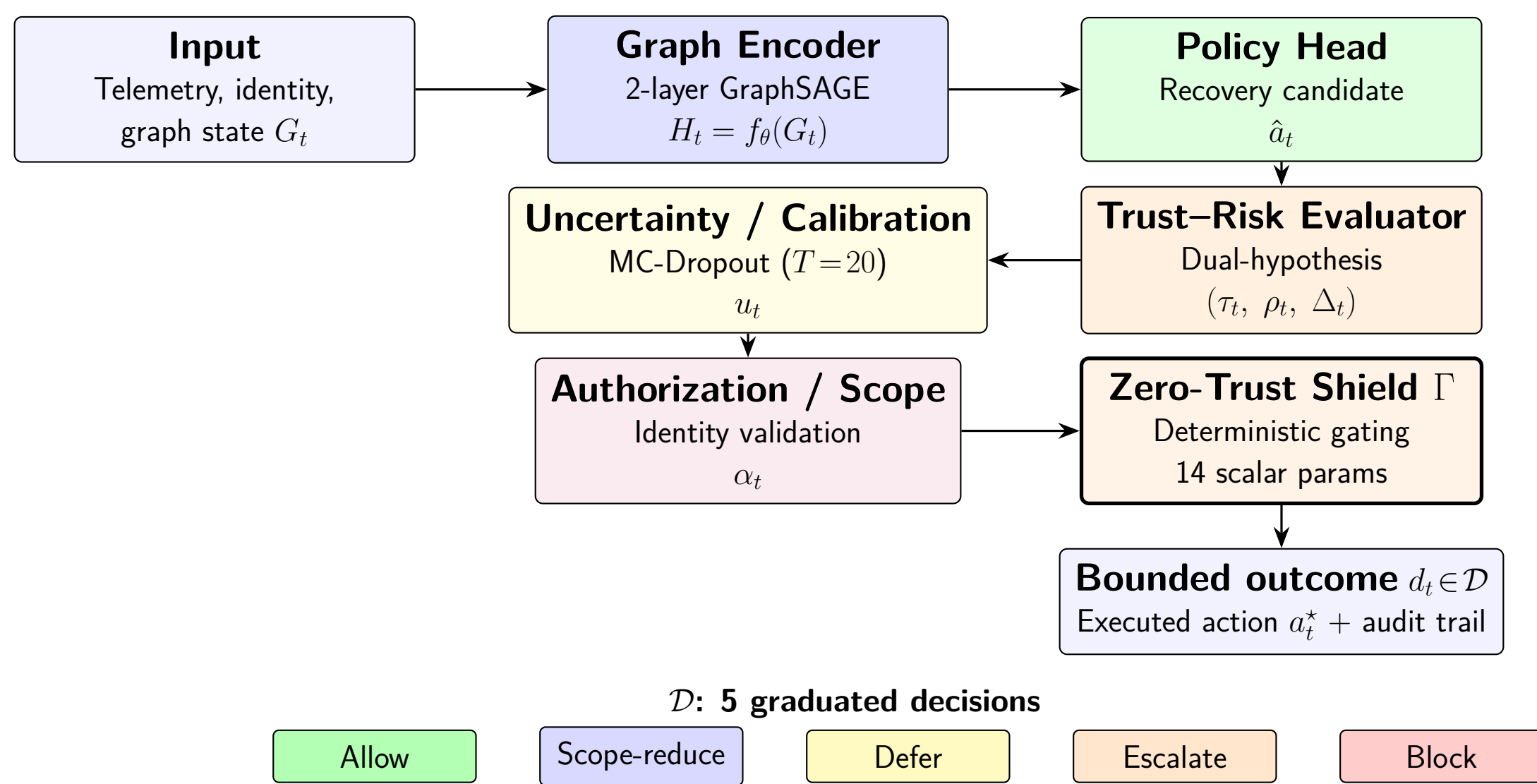
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- Key insight:** Embodied carbon dominates operational by $10^6 \times 1$. Maximize useful packets to amortize.
- CDL Pareto bound:** provable convergence to near-optimal PDR *and* bounded carbon budget.

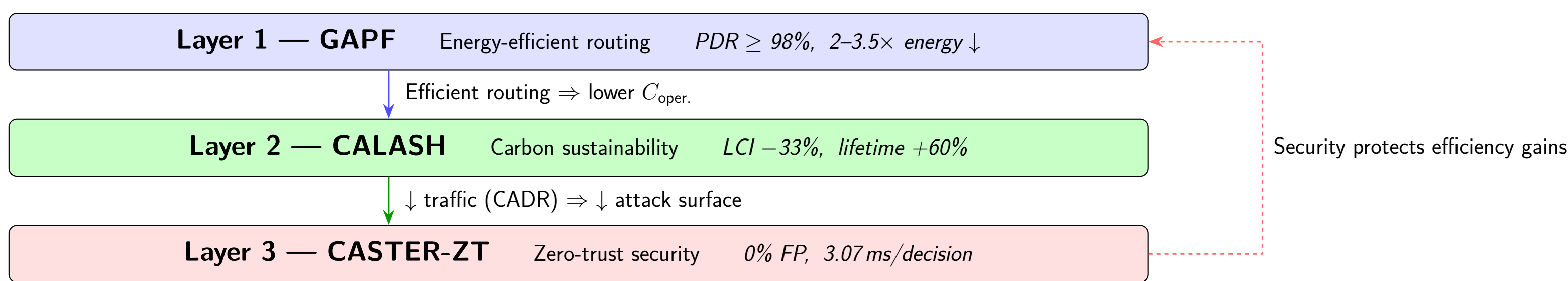
Article 3: CASTER-ZT — Zero-Trust Shielded Decision Control

IEEE Trans. Mobile Computing



- 10 formal propositions:** monotone conservatism, unauthorized-actuation exclusion, conformal coverage.
- NIST SP 800-207** compliant; shield is **explainable** and **auditable**.
- “Policy proposes, shield disposes” — security-first design.

Protocol Stack Synergies



Full stack: **~27K parameters** $\approx$ 1 RNN agent, $\sim$ 103 KB $\Rightarrow$ **Cortex-M7 deployable**.

Gap Analysis: 12 Design Criteria

Criterion	LEACH	QMIX	CPO	Our Stack
Energy-efficient routing	✓	✓		✓
Topology awareness (GNN)				✓
MARL policy fusion				✓
Pareto-monotonic scalarization				✓
Lifecycle carbon metric (LCI)				✓
Carbon-aware data reduction				✓
Carbon budget guarantees				✓
Post-disaster self-healing				✓
6G physical layer				✓
Rogue policy detection			✓	✓
Graduated decisions (5 levels)				✓
Formal security guarantees				✓
Total (/12)	1	1	1	12

TinyML Deployability

Framework	Params	Memory	Target	Compression
GAPF	1,473	5.8 KB	Cortex-M4	None needed
CALASH	~5K	19.5 KB	Cortex-M4	INT8 ($\rightarrow$ 4.9 KB)
CASTER-ZT	~20K	78 KB	Cortex-M7	Distill.+INT8 ($\rightarrow$ 20 KB)
Full stack	~27K	103 KB	Cortex-M7	Mixed quant.

MCUNet deploys $\sim$ 1M params on Cortex-M7 — our stack is **38 $\times$ lighter**.

Quantitative Results Summary

Metric	Best baseline	Baseline	Ours	Gain
PDR	QMIX	46–67%	$\geq$ 98%	$+43\text{--}111\%$ $\uparrow$
Energy / pkt	QMIX	$1.0\times$	$0.29\text{--}0.50\times$	$2\text{--}3.5\times$ $\downarrow$
Parameters	RNN agent	$\sim$ 39K	1,473	$26\times$ $\downarrow$
Training memory	QMIX	3.2–13.8 GB	951 MB	$14.5\times$ $\downarrow$
Network lifetime	LEACH	931 rounds	1,486	$+60\%$
LCI (CO ₂ /pkt)	LEACH	13.42	8.99	-33%
Rogue det. (0% FP)	Best 0-FP	0%	74–98%	$+74\text{--}98$ pts
False positives	Shield-Binary	48.8%	0%	Full elim.
Latency / decision	O-RAN budget	10 ms	3.07 ms	$3.3\times$ $\downarrow$
Criteria (/12)	Best existing	1	12	$12\times$ $\uparrow$
Total test runs				$>$ 23,800

GAPF: Packet Delivery Ratio (7 Scenarios)

<i>n</i>	<i>R</i>	QMIX	QTRAN	QPSOFL	GAPF
20	25	.559	.558	.075	.753
30	25	.568	.567	.053	.877
50	35	.553	.552	.153	.990
70	35	.545	.545	.122	.980
70	45	.667	.659	.365	.980
100	35	.464	.462	.099	.980
100	50	.669	.684	.448	.980

3 seeds $\times$ 1,000 episodes = **21,000 test episodes** across 20–100 sensors.

Six Original Scientific Contributions

- C1** First **MARL policy fusion** via **GNN** for WSN routing (GCN+CAS, 1,473 params).
- C2** **Monotonic Q-K Attention Network** with formal Pareto monotonicity guarantee.
- C3** First **Lifecycle Carbon Intensity (LCI)** metric for WSNs (ISO 14040).
- C4** First **4-pillar carbon-aware protocol** with Lyapunov convergence guarantees.
- C5** First **zero-trust decision control** for AI-native networks (10 propositions, 0% FP).
- C6** Demonstration of **complementarity**: efficiency $\rightarrow$ sustainability $\rightarrow$ security.

SDG Alignment and Impact

SDG	Target	Contribution
9	Resilient infrastructure	GAPF: 1,473 params on Cortex-M4
11	Reduce disaster losses	$\text{PDR} \geq 98\%$, SHDR recovery
12	Sustainability reporting	LCI: ISO 14040 metric
13	Climate resilience	CADR: carbon-adaptive sensing
16	Transparent institutions	Shield: 0 FP, auditable

Conclusion and Future Work

This thesis demonstrates that AI, designed with **parsimony** ($\sim$ 27K params), **environmental awareness** (LCI), and **systematic verification** (zero-trust), is a reliable tool for critical environments.

Future Directions

- Short (1–2 yr):** Unified stack integration; hardware testbed (Zolertia, Coral, Jetson).
- Medium (2–4 yr):** Federated learning; O-RAN digital twins; formal certification (Coq/Isabelle).
- Long (4+ yr):** Non-terrestrial networks (LEO/HAPS); LLM meta-controllers; standardization.

Transferability: Precision agriculture | Smart cities | Connected health | Drone swarms.

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